Histology of the Cardiovascular System

Learning outcomes: By the end of this lecture and following completion of the self study you should be able to:

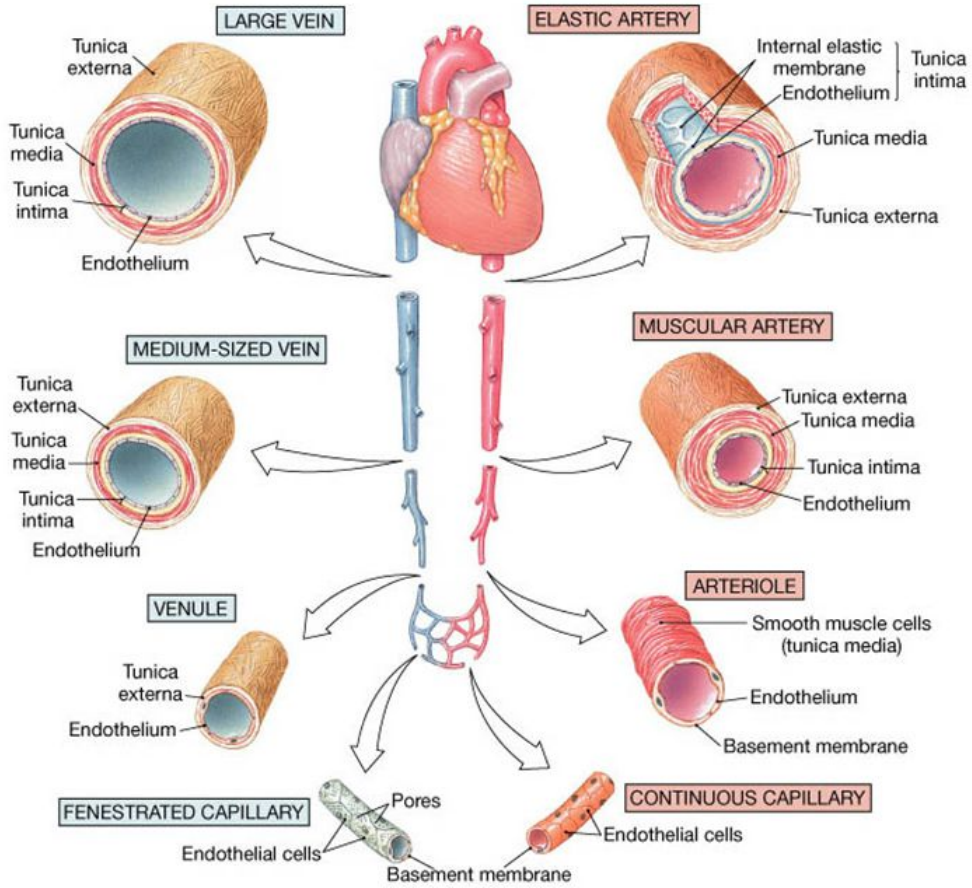
- describe how blood vessels (arteries, arterioles, capillaries, venules and veins) are named.
- describe the structure of different types of blood vessels in relation to their function in supplying blood to and from the tissues of the body.

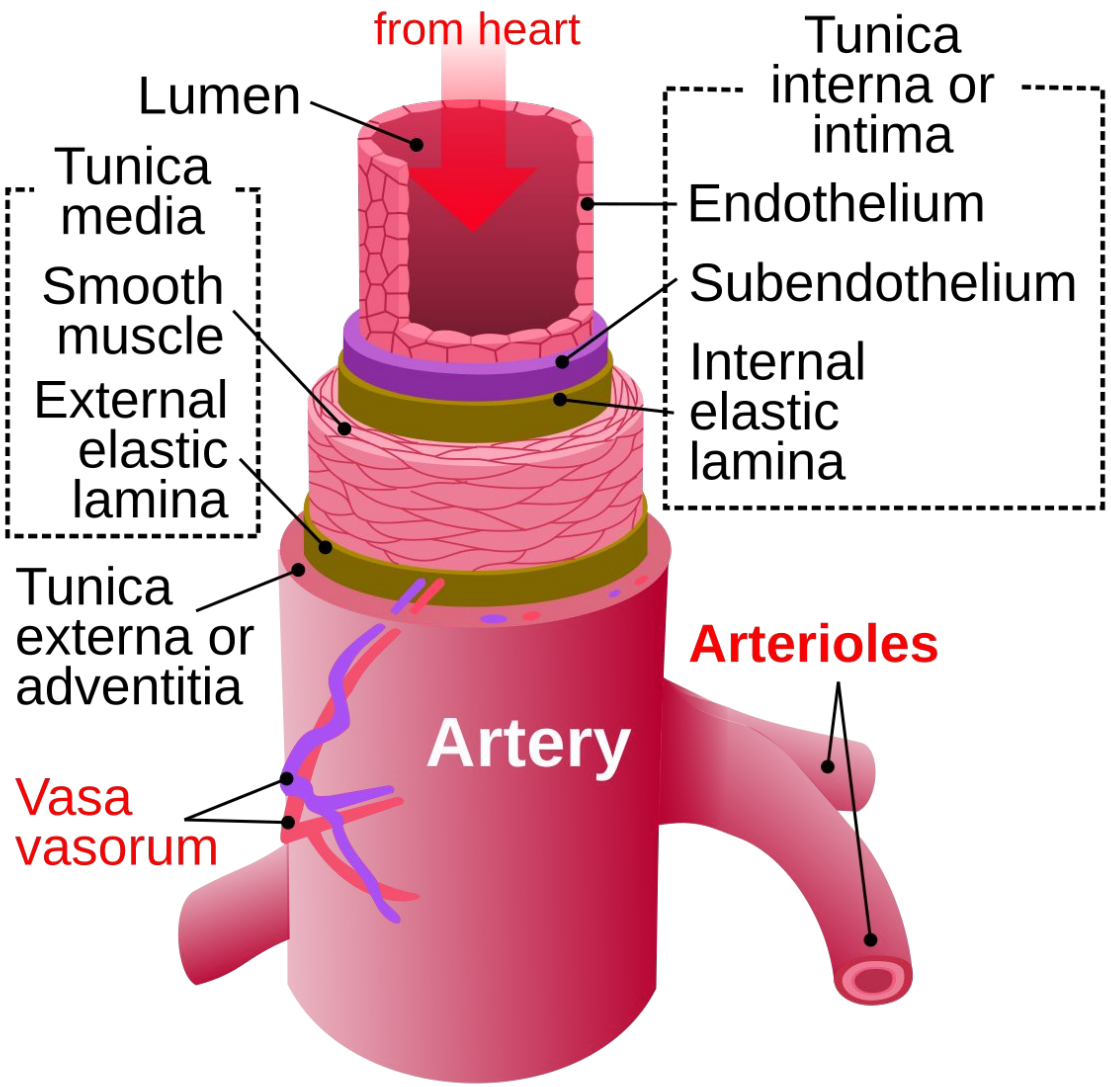
Types of Blood Vessels :

- **Arteries** are blood vessels that carry blood away from the heart to supply the organs and tissues of the body. Different arteries contain varying amounts of elastic fibres and smooth muscle fibres in their walls; thus they are named **elastic (conducting)** and **muscular (distributing)** arteries. The walls of the elastic arteries expand slightly with each heartbeat.

- The muscular arteries branch into **arterioles** whose function is to **regulate the amount of blood reaching an organ or tissue and more generally in regulating blood pressure.** The diameter of the muscular arteries and arterioles is controlled by the **autonomic nervous system**. The arterioles branch into smaller vessels (**metarterioles**), which carry blood into the smallest vessels in the body, the **capillaries.**

Blood vessel types and functions

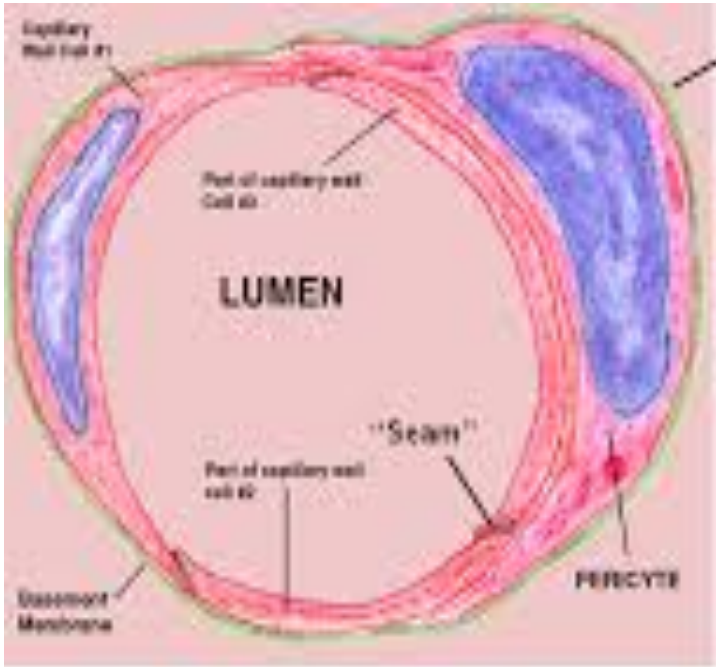



from heart
Lumen
Tunica media
Smooth muscle
External elastic lamina
Tunica externa or adventitia
Vasa vasorum
Tunica interna or intima
Endothelium
Subendothelium
Internal elastic lamina
Arterioles
Artery

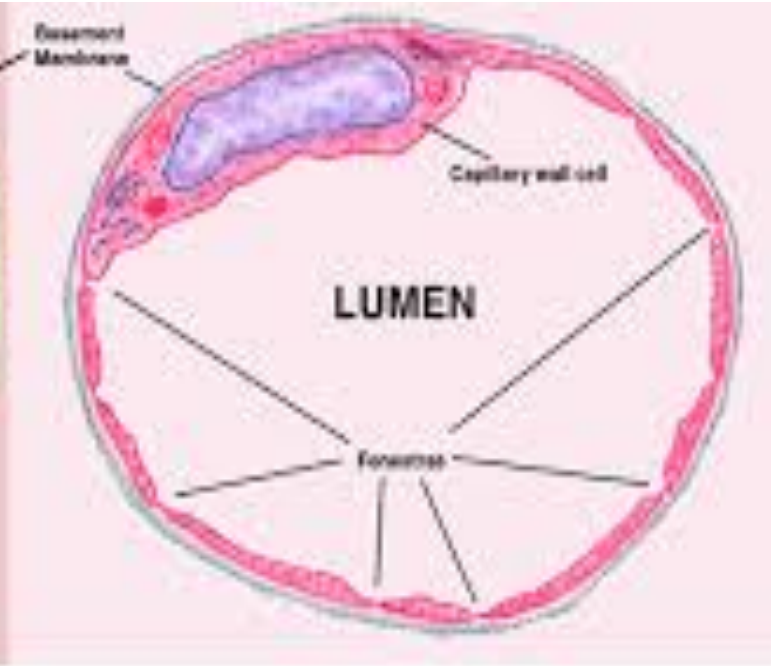
- A capillary wall is mostly one cell thick and allows exchange of substances between blood and tissues.

- **The capillary wall may be continuous or fenestrated**. Both these types of the capillaries may be surrounded partially by **pericytes**.

- In addition to these two types of capillaries, there is a category of vessels found in (**the liver, spleen and the bone marrow**), called **sinusoids**, which are generally larger in diameter and may contain special lining cells and an incomplete basal lamina. Under certain conditions, some blood cells leave the circulatory system to enter the tissue spaces.



Closed or Continuous Capillary



Fenestrated Capillary

- Capillaries merge into large vessels called the **venules** which merge to form even larger vessels called the **veins** which carry blood towards the heart.

- The construction of a vein is essentially similar to that of an artery, except that its wall is thinner and its lumen wider and irregular.

- The veins usually contain **semilunar paired valves** that permit blood to flow in only one direction.

- Those veins are narrower than 1 mm in diameter and **those in the thoracic and abdominal cavities do not have valves**.

- The veins collapse if blood pressure is not maintained.

- The blood flow in arteries is the result of cardiac systolic pressure, whereas blood flow in veins is, to a great extent, determined by the "**muscle-pump**" action in the leg and pressure factors in the abdominal and the thoracic cavities.

- It is important to realise that blood vessels reach almost every part of the body except (**cartilage, epithelia, cornea and a few other structures**).

THANK YOU